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Monitoring IMS Log-Based Change Data Capture

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Abstract

This article describes how to use PowerExchange messages that report IMS change capture counts to monitor IMS log-based ECCR change capture activity.

Supported Versions

- PowerExchange 10.0 or later

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Overview

To determine the level of IMS log-based ECCR change capture activity, view the PWX-06152 and PWX-01653 messages that PowerExchange logs to the ECCR message data set. This data set is specified by the DTLLOG DD or DTLLOG*nn* DD in the IMS log-based ECCR JCL.

For PowerExchange to generate these messages, you must specify CAPT_STATS=Y in the RUNLIB(CAPTIMS) configuration member for the IMS log-based ECCR.

The IMS log-based ECCR captures change data from IMS system log data sets (SLDSs) that are associated with the IMS online region, IMS DB control region, or IMS batch DL/I jobs. Each time the ECCR completes processing an SLDS and closes it, PowerExchange generates a PWX-06152 message and a corresponding set of PWX-06153 messages that report the following statistics by IMS database name and user-defined registration name for segments of interest:

- Number of inserts captured
- Number of deletes captured
- Number of replacements captured

These capture counts indicate the change capture activity since the last time PowerExchange issued a PWX-06152 message. These counts are for a specific SLDS, the last one closed, if the start and end timestamps of the change records in the SLDS do not overlap with the timestamps of the records in another SLDS.

If two or more SLDSs contain change records with overlapping timestamps, the ECCR opens all of these SLDSs and processes them concurrently. The ECCR logs the changes to the PowerExchange Logger log files in chronological sequence by timestamp. When the ECCR completes processing one of these SLDSs and closes it, PowerExchange generates a PWX-06152 message and a set of PWX-06153 messages that report the number of changes that the ECCR read from all of the SLDSs up to the current point. When the ECCR closes another SLDS, PowerExchange issues another PWX-6152 message and reports change activity since the previous PWX-06152 message. This way, PowerExchange reports a change only once in the capture counts for SLDSs with overlapping timestamps.

Example Message Output

PowerExchange writes the following messages to the DTLLOG DD for a specific SLDS named ABC1.IMS.LOGBASED.G0001V00. The messages indicate the change capture activity since the last time a PWX-06152 message was issued. The PWX-06153 messages presents change capture counts for registration names that are associcated with six IMS databases, for example, registration name "active.1" for database "FDPVF2."

```
PWX-06152 IMS SLDS File ABC1.IMS.LOGBASED.G0001V00  closed with some records processed.
Progress since last statistics display:
PWX-06153 activ.1         FDPVF2          ISRT=0000000020 DLET=0000000002 REPL=0000000003
```

```
PWX-06153 address.1       HALADDR         ISRT=0000000015 DLET=0000000005 REPL=0000000005
PWX-06153 baa200.1        DBLOGBOF        ISRT=0000000002 DLET=0000000000 REPL=0000000001
PWX-06153 bba200.1        DBLOGBOF        ISRT=0000000010 DLET=0000000001 REPL=0000000001
PWX-06153 bbb200.1        DBLOGBOF        ISRT=0000000009 DLET=0000000002 REPL=0000000004
PWX-06153 bca200.1        DBLOGBOF        ISRT=0000000000 DLET=0000000000 REPL=0000000000
PWX-06153 caa400.1        DBLOGCOF        ISRT=0000000012 DLET=0000000003 REPL=0000000010
PWX-06153 cba400.1        DBLOGCOF        ISRT=0000000006 DLET=0000000000 REPL=0000000001
PWX-06153 cbb400.1        DBLOGCOF        ISRT=0000000007 DLET=0000000001 REPL=0000000001
PWX-06153 cca400.1        DBLOGCOF        ISRT=0000000008 DLET=0000000000 REPL=0000000003
PWX-06153 cfcac01.1       CUSTOMER        ISRT=0000000004 DLET=0000000002 REPL=0000000005
PWX-06153 cfcac02.1       CUSTOMER        ISRT=0000000020 DLET=0000000005 REPL=0000000009
PWX-06153 cfroot.1        CUSTOMER        ISRT=0000000050 DLET=0000000009 REPL=0000000012
PWX-06153 daa01k.1        DBLOGDOF        ISRT=0000000006 DLET=0000000000 REPL=0000000004
PWX-06153 dba01k.1        DBLOGDOF        ISRT=0000000023 DLET=0000000008 REPL=0000000013
```

Note: To find the capture counts for a particular segment, first look up its registration name in the PowerExchange Navigator.

Author

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